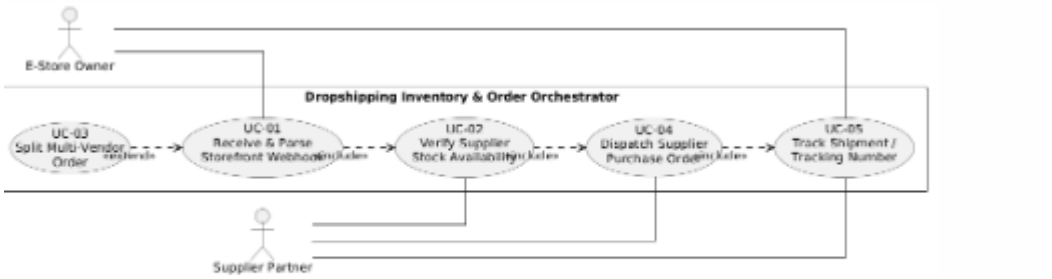
SOFTWARE ENGINEERING

LAB 1

Name: BR MANASI

Section: B
SRN:PES2UG24CS104

Use case flow diagram



Requirements table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall parse inbound Shopify/WooCommerce order webhooks, verify supplier API inventory, and auto-dispatch supplier purchase orders.	High	Pass: Supplier PO triggered with customer delivery address. Fail: Order placed for out-of-stock supplier.	Ensures orders are only placed against suppliers with verified stock, preventing broken fulfillment promises to the end customer.
FR-002	Functional	The system shall split a multi-vendor customer order into separate supplier purchase orders, one per supplier whose SKUs appear in the order.	High	When an order contains items from 2+ suppliers, the system generates one PO per supplier with the correct line items; across a sample of 20 multi-vendor test orders, each PO's items sum back to the original order with no duplication or omission.	Since inventory is sourced from independent Supplier Partners, correct splitting is essential to avoid misrouted or missing shipments.
FR-003	Functional	The system shall allow a Supplier Partner to submit a tracking number against a dispatched purchase order.	High	When a Supplier Partner submits a valid tracking number for a PO in "Dispatched" status, the PO status changes to "Shipped" and the	Tracking numbers are the mechanism by which the E-Store Owner can monitor multi-vendor fulfillment.

				number is stored within 5 seconds; submission is rejected if the PO is not yet "Dispatched."	
FR-004	Functional	The system shall allow the E-Store Owner to view the current status of customer orders, supplier purchase orders, inventory verification results, and shipment tracking information.	Medium	When the E-Store Owner opens the order dashboard, each order displays its associated supplier PO(s), latest stock-verification status, and tracking status when available, with updates reflected within 10 seconds of a status change	Provides centralized visibility into the multi-vendor fulfillment process and helps the E-Store Owner identify stock or shipment issues.
FR-005	Functional	The system shall automatically retry supplier inventory verification when a supplier API does not return a valid stock response.	Medium	When an inventory check fails to return a valid response, the system retries the request up to 3 times. If all retries fail, the affected order line is marked “Inventory Verification Failed” and the E-Store Owner is notified.	Improves reliability of the real-time inventory verification process and reduces failures caused by temporary supplier API issues..
NFR-001	Nonfunctional—Performance & Security	Inbound webhook processing must acknowledge receipt with HTTP 200 within 200 ms to prevent provider webhook retry storms.	High	Pass: load testing at 500 webhooks/second confirms a 200 OK response within 200 ms for at least 99% of requests. Fail: latency exceeds 200 ms or success rate drops below 99%, risking a retry storm.	Storefront platforms automatically resend webhooks on slow/missing acknowledgment; failing to respond fast enough causes duplicate order processing..
NFR-002	Nonfunctional—Security	The system shall authenticate all inbound storefront webhooks and outbound supplier API calls using signed requests (e.g., HMAC signatures), rejecting any request with an invalid or missing signature.	High	Penetration testing confirms 100% of unsigned or invalidly-signed webhook/API requests are rejected with an appropriate error, and no order or purchase order is ever created from an unauthenticated request..	The system auto-dispatches real purchase orders on the store owner's behalf; a forged webhook or spoofed supplier response could trigger fraudulent orders or fake stock data.

Use-Case Flow: Ingest Order & Dispatch Supplier Purchase Orders

Actors: Storefront Platform , Supplier Partner , E-Store Owner

Preconditions:

- The E-Store Owner's storefront webhook endpoint is registered and active.
- Supplier Partners' inventory APIs are reachable and authenticated.
- A customer has completed checkout on the storefront, generating an order.

Postconditions:

- *Success:* One purchase order per involved supplier is dispatched in "Dispatched" status, each linked back to the original order.
- *Failure (partial):* Any line item that remains out-of-stock after retries is marked "Cancelled - Out of Stock," while in-stock lines from other suppliers still proceed to dispatch.

Main Success Scenario

1. Storefront Platform sends an order webhook containing the customer's order details and line items.
2. System acknowledges receipt with HTTP 200 within 200 ms (per NFR-001).
3. System parses the payload and identifies the supplier SKU for each line item.
4. System groups line items by supplier and queries each relevant Supplier Partner's inventory API in real time *(include: Verify supplier inventory)*.
5. System confirms stock availability for all line items across all suppliers.
6. System splits the order into one purchase order per supplier *(include: Split multi-vendor order)*.
7. System auto-dispatches each purchase order to its corresponding Supplier Partner, including the customer's delivery address.
8. System updates each PO's status to "Dispatched" and links it to the original order.
9. System notifies the E-Store Owner that the order was successfully processed and dispatched.

Alternate Flow — 5a. Supplier SKU Out of Stock

5a1. The inventory check for one or more line items returns "out of stock."

5a2. System retries that inventory check up to 3 times at defined interval.

5a3. If still out-of-stock after 3 retries, the system cancels that order line, marking it "Cancelled - Out of Stock."

5a4. System notifies the E-Store Owner of the cancellation within 1 minute of the final retry.

5a5. Processing resumes at main step 6 for any remaining in-stock line items, so unaffected suppliers still receive their purchase orders.